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COI Name (Institution):	Burma, Sandeep (The University of Texas Southwestern Medical Center) Chen, David (The University of Texas Southwestern Medical Center) Gazdar, Adi (The University of Texas Southwestern Medical Center) Girard, Luc (The University of Texas Southwestern Medical Center) Richardson, James (The University of Texas Southwestern Medical Center) Shay, Jerry (The University of Texas Southwestern Medical Center) Story, Michael (The University of Texas Southwestern Medical Center) Wright, Woody (The University of Texas Southwestern Medical Center) Xie, Xian-Jin (The University of Texas Southwestern Medical Center) Minna, John (The University of Texas Southwestern Medical Center) Yang, Chin-Reng (The University of Texas Southwestern Medical Center)		
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Task Description:	The University of Texas Southwestern Medical Center (UTSW) NSCOR focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes are being compared in these models to that of gamma-irradiation. These dose risk assessments are designed to measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. They also measure effects on development of cancer "stem-like" cells ("cancer initiating cells"). 3D organotypic culture and animal models also allow measurement and risk estimation in both tissue surrogates and in vivo lung tissues. These individual risk estimates can then be combined into a model (such as a two-stage clonal expansion model) for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORs and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have completed many studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified, that specific expression profiles for HZE irradiation exist, and that HBECs genetically manipulated to progress part of the way toward malignancy are even more sensitive to HZE-induced functional changes. In addition, we are completing experiments testing whether transgenic mouse models of lung cancer after low and high LET exposure increase the early onset or progression of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation; 2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells; 3. Effect of HZE particle irradiation on functional progression of human lung cancer at the cellular and organotypic level; and 4. Effects of HZE particles on the development of lung cancer in vivo in novel mouse models. These projects are supported by 4 Cores: Administrative; Cell Culture; Genomics and Proteomics; and Biostatistics and Bioinformatics. In addition, with funding from the Department of Energy (DOE) contributing to this NSCOR, studies of low-LET and low dose radiation in the same model systems are also being undertaken. The HZE particle irradiation is done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer, a Center for the NCI Early Detection Research Network (EDRN) program, and is part of the NCI sponsored Genetic Epidemiology of Lung Cancer Consortium (GELCC) there is great synergism and additional resources available for the successful completion of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused by mostly by smoking exposure, a significant number of lung cancers arise in life time never smokers and thus is a sizable health problem in the USA. Indeed lung cancer in never smokers is the 7th leading cause of world-wide cancer related deaths. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high-LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. Over the past 10 years there has been a resurgence of interest in the use of computed tomography (CT) scanning for the early detection of lung cancer. Currently we are awaiting the results of the National Cancer Institute (NCI) National Lung Cancer Screening Trial (NLST) which randomized ~55,000 current and former smokers to be screened regularly with CT scans on a yearly basis or not to determine if such screening would impact lung cancer mortality. While there is a lot of debate about this area, this trial should resolve this important issue. If the decision is to use CT scans for screening in the general population there will be a large increase in the exposure of potentially individuals at very high risk of developing lung cancer (current and former smokers) to repeated doses of radiation from the CT scans. This falls into to the area of low dose radiation exposure covered by the Department of Energy (DOE) supported part of this application. Thus, our proposed low dose and low dose rate studies of gamma radiation in the HBEC and mouse models of lung cancer Projects are of great immediate relevance to developing risk models for such radiation exposure in the USA. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 45 different individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as expression of oncogenic K-RASv12, knockdown of expression of mutant p53, expression of oncogenic mutant EGFR, expression of c-myc). Our studies are providing quantitative data following high-LET irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high-LET radiation in space in order to reduce the uncertainties in prediction of risk of astronauts developing lung cancer as part of long-term missions in space However, the results will also have major impact and benefit to life on Earth includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation which are of potential importance as markers of radiation exposure that could occur through environmental exposures, accidentally or through terrorism. As part of this will be the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional cultures, and finally growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are also learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA and protein expression, and signaling pathway changes. Our large HBEC panel coupled with new high density single nucleotide polymorphism (SNP) analysis will allow describing polymorphic differences in such

responses.

Our lung cancer NSCOR is addressing the following specific critical path roadmap and gap questions:

•We are improving the understanding of space radiation in lung cancer initiation, promotion and progression spanning cell and molecular biology at the cell, tissue and system biology level that will lead to significant reduction in the uncertainties in risk projection models

•We are investigating the increased risk from space radiation as a function of age of exposure, age, latency, gender, tissue, radiation quality and dose rate?

•We are developing tissue specific risk models using human 3D culture and animal models for lung cancer

•We are investigating how aberrant DNA damage processing, genomic instability, epigenetic effects including methylation, persistent oxidative damage, altered senescence, and non-targeted effects contribute to irradiation-associated lung cancer using a variety of surrogate endpoints that are being validated.

Overall we are conducting research to investigate how biological endpoint and systems biology approaches can be used to integrate research on the cellular and molecular as well as tissue level of radiation damage that will lead to significant reductions in the uncertainties in risk projection models of lung cancer. As will be described in this report, we are confident and will provide compelling evidence that the sum of the efforts of our assembled team are significantly contributing more to assessing cancer risk associated with space radiation compared to the individual components.

Project 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation. Michael Story, Ph.D. Lead, Adi F. Gazdar, M.D. Co-Lead, John D. Minna, M.D. Co-Lead.
Specific Aim 1: To quantitate the effects of HZE particle irradiation in causing genetic changes in HBECs and their isogenic derivates.

Specific Aim 2. To quantitate the amount of epigenetic changes caused by HZE particle irradiation in HBECs and their isogenic derivates using high throughput quantitative methylation specific PCR (MSP) assays for acquired promoter methylation

Specific Aim 3. To quantitate the effects of HZE particle irradiation in causing expression profile (mRNA and protein) changes in human bronchial epithelial cells (HBECs) and their isogenic derivates containing various lung cancer-related mutations using array based genome wide and multi protein detecting approaches.

Specific Aim 4. To determine the effect of inter-individual variation on the effects of HZE particle irradiation in causing genetic changes, epigenetic changes, mRNA and protein expression profile changes in human bronchial epithelial cells (HBECs).

Project 1: In the last year this project has focused on 3 specific areas. mRNA Expression Analysis. The first area is comparative gene (mRNA) expression analysis at equi-toxic doses of low LET radiation vs that of HZE radiations, specifically 56Fe. DNA Methylation Profiles. The second area of emphasis this last year was the analysis of radiation-induced changes in DNA methylation patterns. miRNA Expression Profiling. The third area of emphasis dovetails with our gene expression analysis, and represents a relatively new area of research for radiation exposures.

Project 2: Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells. David Chen, Ph.D. Project Lead, Sandeep Burma, Ph.D. Co-Lead.

Specific Objectives :

Aim 1. To test the hypothesis that the immediate cellular response to DNA damage induced by HZE would be different and more complex than that induced by X-rays.

Aim 2. To test the hypothesis that persistent DNA damage induced by HZE is more complex and difficult to repair than that induced by X-rays and, therefore, more deleterious.

Aim 3. To determine if damage induced by HZE is preferentially repaired by non homologous end joining (NHEJ) or homologous recombination (HR) or is preferentially repaired in certain phases of the cell cycle.

Aim 4. To compare the response of human bronchial epithelial cells (HBECs) as non immortalized, immortalized, and with premalignant oncogenic changes in 3D cultures to HZE exposure using a high throughput approach.

Summary of Current Progress/Detailed Results

1. Damage Responses of HBECs Irradiated with Fe Particles.

2. Repair kinetics of DNA damage induced by Fe particles in HBECs

3. DNA damage responses in organotypic 3D cultures of HBECs.

4. Establishment of a YFP-53BP1 system for live cell imaging of DNA damage and its repair:

5. Recruitment of YFP-53BP1 to the sites of DNA DSB in living cells is dose-dependent:

6. Live cell imaging of repair of DNA DSBs induced by gamma-radiation:

7. Expression of EGFP-53BP1 in HBEC3KT cells:

8. EGFP-53BP1 forms foci in 3D cultures of HBEC3KT cells.

9. To compare the ability of monolayer and 3D cultures of HBECs to carry out repair of DSBs induced by gamma rays or HZE by live cell imaging.

Project 3. Effect of HZE Particle Irradiation on Functional Progression of Human Lung Cancer. John D. Minna, M.D. Project Lead, Jerry Shay, Ph.D. Co-Lead.

Task Progress:

Specific Aim 1. Effects of HZE particles on early events in lung cancer progression (soft agar colony formation and other markers of cellular transformation).

Specific Aim 2. Effects of HZE particles on intermediate events in lung cancer progression (invasion in 3 D organotypic cultures).

Specific Aim 3. Effects of HZE particles on late events in lung cancer progression (ability to form tumors in immune deprived mice).

Immediate and long-term plans:

Project 1 is now very close to hitting the milestones described in the original application. We will very shortly have a set of well characterized genomic responses (both gene and miRNA expression) to low LET and HZE radiation exposures in a series of isogenic immortalized human bronchial epithelial cells (HBECs) which differ only by specific oncogenic changes introduced (e.g. KRAS, p53, EGFR, E6, E7). These variants thus include cells with altered p53, RAS and EGFR signaling. Our goal will be to mine the data available in order to develop biomarkers of response that may be used for relative risk determinations of lung cancer from space radiation exposure. Determining such biomarkers dovetails with the transformation studies of Project 3. We will use multiple analysis methodologies to develop biomarkers including Gene Set Enrichment, pathway analysis, smar, and in particular, Bayesian analysis for interaction nodes. In future studies these biomarkers can be evaluated in mouse models and in particular, can be used to address inter-individual differences in carcinogenic response for risk determination.

Project 4. Effects of HZE Particles on the Development of Lung Cancer in Novel Mouse Models. Jerry W. Shay, Ph.D. Project Lead, James Richardson, DVM, Ph.D. Co-Investigator, Oliver Delgado, Student Assistant.

Specific Aim 1. Use the K-RAS transgenic mouse model of human lung cancer to determine effects of low- and high-LET radiation in tumor initiation and progression.

Specific Aim 2. Use additional transgenic mouse models to determine the effects of low- and high-LET radiation in both living animals and in isolated tissues.

Specific Aim 1. Use the K-RAS transgenic mouse model of human lung cancer to determine effects of low- and high-LET radiation in tumor initiation and progression.

Summary:

1. Fractionated and single dose irradiation of 1Gy 56Fe- particles do not increase tumor incidence

2. Tumors in the LA1 K-RAS mice irradiated with fractionated doses of (0.2Gy/day, total 1 Gy (1GeV/n) 56Fe-particles have a statistically significant increased probability of progressing to invasive carcinoma in comparison to a single acute 1Gy 56Fe- dose.

3. These results suggest significant effects of fractionated high-LET irradiation on cancer progression

4. To test this hypothesis we have initiated additional single dose and fractionated doses to confirm these important observations. We have irradiated mice with fractionated x-ray (0.6 Gy/day x 5 days) and have not observed an increase incidence of invasive tumors. We have initiated fractionated 56Fe- particle experiments (0.1 Gy/day x 10 over a 12 day period).

5. We are testing the hypothesis that fractionated high-LET irradiation induces persistent inflammatory signaling that does not occur in single doses. An alternative hypothesis is that there could be a role of stem cell repopulation for lung tissue that is important in cancer progression. Thus, acute doses of high-LET ions may increase cell killing, reducing the number of target stem cells available for initiation and progression to lung cancer.

6. We have provided non lung tissues from our control and irradiated mice to other investigators (Ameila Eisch, brain tissue; Zhi-Min Yuan, kidney, liver, spleen)

Core A: Administrative Core. Leads: John Minna and Jerry Shay, and UTSW NSCOR

Administrator: Brenda Zielke.

The Administrative Core has established accounts and control over disbursement of all funds from this NSCOR application. It has also kept track of all personnel employed by this NSCOR. It has provided a central place for communication and coordinated data sharing In addition, it has established a series of biweekly to monthly scientific and planning meeting of the NSCOR investigators at UTSW. Approximately 25 scientific personnel attend each of these meetings. At each meeting administrative matters are discussed, experiments are planned, administrative changes made by the senior leadership team, and most importantly a scientific presentation by one of the NSCOR investigators is made and critiqued by the other investigators. Dr. Minna serves as the Administrative Core Lead with Dr. Jerry Shay as the Administrative Core Co-Lead. The Administrative Core coordinated all of the arrangements for the participation of several NSCOR investigators (Minna, Shay, Chen, Story, Roig, Burma, Delgado) in the 19th NSCOR Investigators Workshop held in Philadelphia, PA June 30-July 2, 2008. The Administrative core also arranged the Internal Scientific Advisory Board Review (March 4, 2008 as part of a mini-retreat at UT Southwestern) and the External Advisory Board Review (October 15, 2008 Mary Helen Barcellos-Hoff, Robert Ullrich and Martin Brown participating).

Core B. Cell Culture. Core Leads: John D. Minna, M.D. and Jerry Shay, Ph.D.

Enhancement of human bronchial epithelial cell 3D lung model of tissue like development There is increasing evidence that cancer arises from stem cell populations that are important for the normal turnover of tissues and the rapid repair of tissues in instances of injury. In the lung there is evidence for a bronchiole alveolar stem cell (BASC) population that may give rise to lung carcinoma. We developed a panel of immortalized human bronchial epithelial cells (HBECs) and have previously demonstrated that these cells when placed in an air-tissue culture medium interface overlying a fibroblast stromal collagen matrix could differentiate into mucous and ciliated bronchial epithelial cells. These results were expected since the cells originated from the central airways and were obtained from mid size bronchial tissues. We examined these cells for a variety of differentiated cell markers and observed that they expressed p63 (a transit amplifying cell marker) and surprisingly surfactant protein A (SP-A), CC10, a Clara cell marker, and when differentiated weak expression of the thyroid transcription factor TITF1 (known to be important in adenocarcinomas of

	the peripheral airway of the lungs). These peripheral markers associated with Clara and type II alveolar cells imply that our immortalized HBECs have enhanced plasticity and should be considered pluripotent lung stem cells. We next placed these cells into matrigel overlying a lung fibroblast (IMR90) feeder layer and observed that the cells could efficiently differentiate into spheroids or cyst like structures. Studies of these HBECs growing in matrigel and forming cysts and SACs indicated they started to express SP-A and morphologically resembled small bronchioles and alveolar sacs. When examined in the electron microscope, these cells also had lamellar bodies which is a hallmark of type II alveolar cells. Core C. Genomics and Proteomics Core. Core Lead: Michael Story, Ph.D., Ching-Reng Yang, Ph.D. Core Co-Lead. Genomics Core: The Core has made two advances in technology that should be described. The first was the analysis of miRNA in the HBEC series. The data collected to date are unique to the radiation community at large. The second area of progress has to do with the generation by Illumina Corp of a new gene expression platform that will drive the price of expression analysis down by half. We are implementing this new platform. This is the second 50% reduction in costs associated with gene expression during the life of this current application. This has already allowed the incorporation of new analytic techniques, miRNA analysis, and with the second cost reduction should allow us the flexibility to expand our analysis further or into new areas of genomics and proteomics. A. Specific Functions of the Genomics Core Microarray expression profiling, genome wide DNA copy number (array CGH), and array based protein expression technologies are maturing. However, they are very precise technologies that requires extensive experience to perform and sophisticated instrumentation to carry out. Furthermore, having a single facility through which all array analysis passes minimizes the variability brought about by different individuals processing samples even though the same platform is being used. For these reasons we include a Core facility (Core C, Genomics and Proteomics) in this NSCOR grant. This NSCOR Core interfaces with the UTSW Cancer Center Microarray and the UTSW school wide Cores. The NSCOR Core has the following duties: 1. Interact with Project Leaders and investigators in the design of experiments that use microarray technologies (mRNA, miRNA, DNA copy number, and proteomics) in their experimental approach. Biostatistics and bioinformatics will also be integrated in this process. 2. Establish procedures for consistency in sample preparation and RNA, DNA, and protein extraction, test all RNA, DNA, and protein samples for integrity, and provide reports to that effect. 3. Perform all hybridizations, perform all scans, and place data on a secure server. 4. Provide early analysis on array quality control and preliminary data analysis to investigators. 5. Interact with the Biostatistics and Bioinformatics Core (Core D) to provide bioinformatic analysis of data according to project requirements. NSCOR Core D: Biostatistics and Bioinformatics. Core Lead: Xian-Jin Xie, Ph.D.; Core Co-Lead: Luc Girard, Ph.D. and Yang Xie, Ph.D. In conjunction with the Genomics (Expression Profiling and Proteomics) Core (Core C) and the Cell Culture Core (Core B), the Biostatistics and Bioinformatics Core (Core D) has the following specific aims: -To provide database support and expertise for the collection, storage, integration, and retrieval of all sorts of data generated from NSCOR projects. -To provide valid and optimal statistical design and to conduct proper analysis required to address the specific aims of each project. -To review background and rationale, to assist in the design, evaluation, and analysis of new research proposal arising from the individual projects. -To assist the other Cores and Projects in the proper analysis and interpretations of different types of bioassays, particularly mega-data such as those generated from gene expression arrays. -To assist in manuscript preparation and to review scientific submissions. -To assist the NSCOR team in building radiation cancer risk models. As we progress on all the NSCOR projects, significant amount of new data have been generated since our 2007 annual progress report. As a result of Core D effort, a comprehensive data management system has been established that allows efficient access and sharing of all NSCOR data. Data Collection Close to 10,000 samples (including cell pellets and cryogenically preserved cell lines) and experiments (irradiation, microarrays, mouse data) have so far been collected and stored in the NSCOR database. These data can be viewed or edited by any participating investigator provided they have the necessary permissions.
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Awards	Camacho C. "008 NASA Space Radiation Summer School, July 2008." Jul-2008
Awards	Camacho C. "Third prize in the poster contest at the 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30-July 2, 2008." Jul-2008
Awards	Gonzalez O. "2007 NASA Space Radiation Summer School, Brookhaven National Laboratory, June 2007." Jul-2007
Awards	Park S. "Travel award to attend 44th Annual Meeting of the Radiation Research Society, Boston, MA, September 2008." Sep-2008
Books/Book Chapters	Xie X-J. "Goodness-of-Fit Tests for Logistic Regression Models." VDM Verlag Publishing, Inc., 2008., Jan-2008